Chromatographic and Electrophoretic Techniques

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Chromatographic and Electrophoretic Techniques (A06312)

Short Title: Chromato. & Electrophor. Tech.
Faculty Science and Computing
Department Science
Cluster Analytical Science
Author DCONNOLLY
Credits 5 **Level:** Introductory

Description of Module / Aims

This module will provide a comprehensive understanding of the theory and practice of a variety of chromatographic and electrophoretic techniques, including thin layer chromatography, liquid chromatography, gas chromatography, and gel electrophoresis. A necessary background will be provided to give the student the ability to select an appropriate separation technique for a given class of analyte.

Programmes

SCIE-0057	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	2 / 4 / M
SCIE-0057	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	2 / 4 / M
SCIE-0057	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 4 / M
SCIE-0057	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	2 / 4 / M
SCIE-0057	BSc in Pharmaceutical Science (WD_SPHSC_D)	2 / 4 / M

Indicative Content

- Basics of solution preparation and associated calculations (concentration, dilutions)
- Basics of intermolecular interactions
- Introduction to chromatography and description of the main sorption mechanisms
- Basic equations of chromatography for the extraction of qualitative and quantitative information from chromatograms
- Principles, practices, apparatus and applications of thin layer chromatography
- Principles, practices, instrumentation and applications of reversed phase and normal phase high performance liquid chromatography
- Principles, practices, instrumentation and applications of gas chromatography
- Principles, practices, instrumentation and applications of alternative modes of liquid chromatography
- Principles, practices, instrumentation and applications of gel electrophoresis
- Associated laboratory-based experiments to supplement and augment the above content

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the difference between molecules based upon their relative polarity.
2. Describe the principles and practices of thin layer chromatography and its applications in the analytical laboratory.
3. Discuss the instrumentation, apparatus and sample preparation required for high performance liquid chromatography, gas chromatography and ion chromatography.
4. Describe the instrumentation, apparatus and sample preparation required for gel and electrophoretic techniques.
5. Describe how the above chromatographic and electrophoretic techniques can be applied to the separation of analytes in a wide range of environments e.g. pharmaceutical, environmental, food, clinical.
6. Operate associated laboratory instrumentation for the analysis of real samples.

Learning and Teaching Methods

- Lectures.
- Practicals.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	50%		1,2,3,4
Continuous Assessment	50%		
Lab Report		40%	6
In-Class Assessment		10%	5,6

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of the learning outcomes. Lack of competence with regards to use of associated laboratory skills, techniques and instrumentation.

40%-49%: Will be able to show limited understanding of the fundamentals of the learning outcomes. Will display some ability with regard to the use of associated laboratory skills, techniques and instrumentation.

50%-59%: As well as the above, the student will be to show reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to use of associated laboratory techniques instrumentation and good laboratory practice.

60%-69%: In addition, the student will demonstrate more detailed knowledge of the all topics contained in the learning outcomes and display good competence with regards to use of associated laboratory instrumentation and very good laboratory practice.

70%-100%: All of the above to excellent levels. In addition the learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to use of associated laboratory techniques and instrumentation and excellent laboratory practice.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Lab	36	
Independent Learning	75	

Essential Material

- Harris, D.C. _Quantitative Chemical Analysis_. 8th ed. New York: W.H. Freeman and Company, 2010.
- Skoog, D.A., D.M. West, F.J. Holler and S.R. Crouch. _Fundamentals of Analytical Chemistry_. 9th ed. California: Brooks/Cole, 2014.

Supplementary Material

- "Waters Chromatography Primers." http://www.waters.com/waters/en_IE/Primers/nav.htm?cid=10048920&alias=Alias_primers_CHEMISTRY_CORPORATE&locale=en_IE